

UNIVERSIDADE DE AVEIRO

PECI – Group 8



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Interactive Application to Support Learning

From Digital Systems to Embedded Systems

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Abstract

The teaching of Digital Systems and Computer Architecture is challenging and presents a high cognitive load, frequently resulting in demotivation due to traditional passive learning methods. This project proposes a distributed and interactive educational ecosystem to mitigate study passivity through gamification and generative artificial intelligence. The mobile application (developed in Flutter) acts strictly as a thin client, delivering micro-learning modules, continuous assessment, and an interactive AI Tutor. To suppress AI hallucinations and ensure pedagogical rigor, the backend relies on a Retrieval-Augmented Generation (strategy to ensure the AI uses specific course bibliography) (RAG) architecture powered by ChromaDB, restricting the Large Language Model (LLM) responses to reliable curricular documents uploaded by professors via web panels (Vue 3). This report details the inception and elaboration phases of the system, defining the requirements and substantiating the adoption of a centralized cloud architecture over local edge deployment.

Acronyms

CU Course Unit

ECI Computer and Informatics Engineering (Engenharia de Computadores e Informática)

LLM Large Language Model

RAG Retrieval-Augmented Generation (strategy to ensure the AI uses specific course bibliography)

XP Experience Points (unit of progression in gamification)

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1. Introduction

1.1 Scope

The system is a cross-platform application (web and mobile compatible) designed to enhance the learning experience in Computer Engineering. Key features include:

- Automatic exercise generation via LLMs.
- Real-time feedback and an intelligent tutoring system.
- A progression system inspired by gamification frameworks (Experience Points (unit of progression in gamification) (XP), streaks, levels).
- Support for both cloud-based AI integration and local/offline execution for resource-constrained environments or privacy-focused use cases.

1.2 Motivation and Purpose

Teaching Digital Systems and Computer Architecture involves high complexity, requiring tools that maintain student engagement. Additionally, the curricular content is often highly dispersed over time, which makes knowledge retention and continuity difficult for many students. The transition to Embedded Systems requires a solid conceptual foundation, which this application facilitates through short, interactive lessons and immediate reinforcement, while also supporting periodic review and self-assessment of previously learned concepts.

This document specifies the functional and non-functional requirements for the "Interactive Learning Support Application", spanning from Digital Systems to Embedded Systems. It aims to provide a technical roadmap for developing a gamified educational platform powered by Artificial Intelligence (AI).

1.3 Objectives

The primary objective of this project is to develop an integrated platform that bridges the gap between theoretical concepts and practical application in Computer Engineering. This includes the implementation of an intelligent tutoring system capable of generating context-aware exercises and providing a seamless offline experience for users in restricted environments.

1.4 Document Structure

This document is organized into chapters covering requirements analysis, system architecture, gamified interface design, and the validation plan within an academic setting.

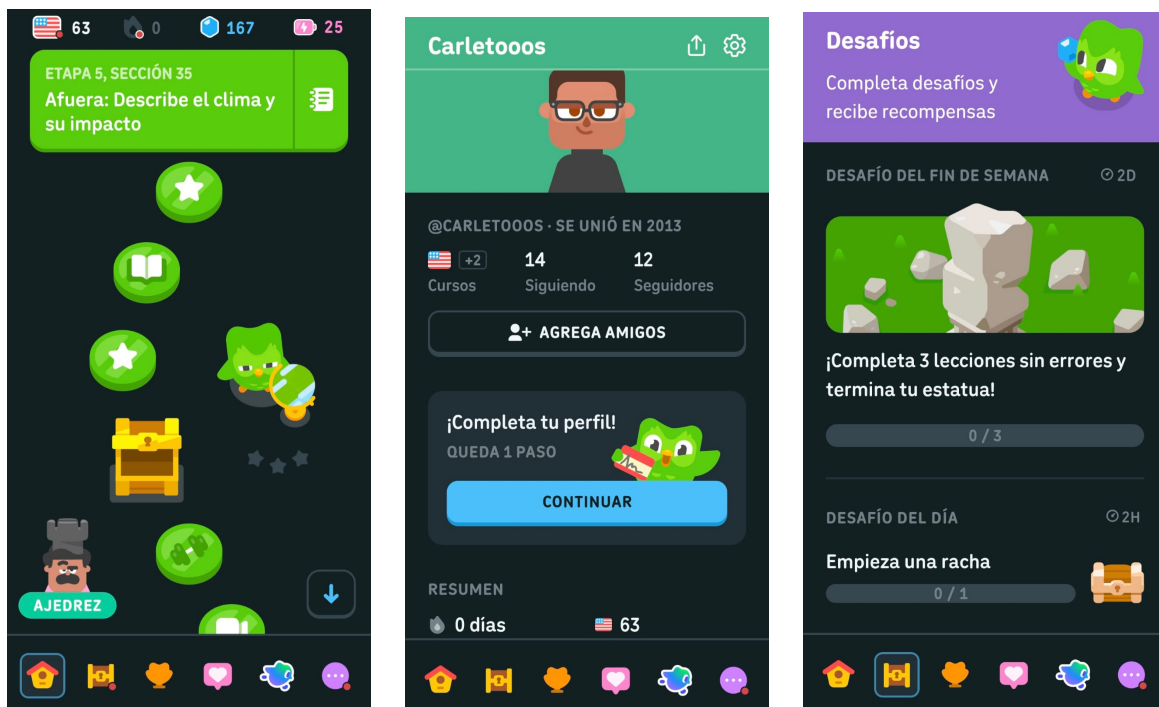
2. State of the Art

2.1 Related Work

The proposed solution establishes a convergence between three distinct educational paradigms: Duolingo, Khan Academy, and NotebookLM. Although these platforms are leaders in their respective fields, they leave a significant structural gap in the specialized domain of Digital and Embedded Systems, where technical complexity is substantially higher.

2.1.1 Duolingo

Duolingo provides the industry standard for gamification and user retention through micro-learning and streak mechanics. It mitigates cognitive overload by atomizing knowledge. This recognition is supported by the findings of Hia et al. [1], which empirically demonstrate that Duolingo significantly improves students' knowledge acquisition compared to traditional methods, thereby validating its effectiveness as a gamification-based learning tool. However, it lacks the technical depth and procedural assessment mechanisms required for higher engineering subjects, operating strictly on static, pre-compiled datasets.



(a) Learning path view, highlighting lesson progression and milestone navigation.

(b) Profile view, showing achievements, streaks, and user progression metrics.

(c) Challenges view, showing task-based practice and completion tracking.

Figure 2.1: Duolingo interface examples used as gamification references.

2.1.2 NotebookLM

Recent advancements in RAG have led to tools like Google NotebookLM, which Reyna [2] identifies as a powerful asset for personalized tutoring. By grounding LLMs in specific course materials, NotebookLM excels in providing accurate, context-aware summaries and

answering student queries while significantly reducing the risk of AI hallucinations.

Nevertheless, NotebookLM functions primarily as a reactive document synthesizer. While it can generate flashcards based on uploaded content, it lacks a structured learning path and a systematic assessment engine. The proposed solution bridges this gap by integrating the grounding capabilities of LLMs with a gamified, curriculum-driven approach. Unlike NotebookLM, which requires the user to manually drive the inquiry, this project implements a proactive pedagogical framework, ensuring students follow a validated progression with continuous, automated feedback loops.

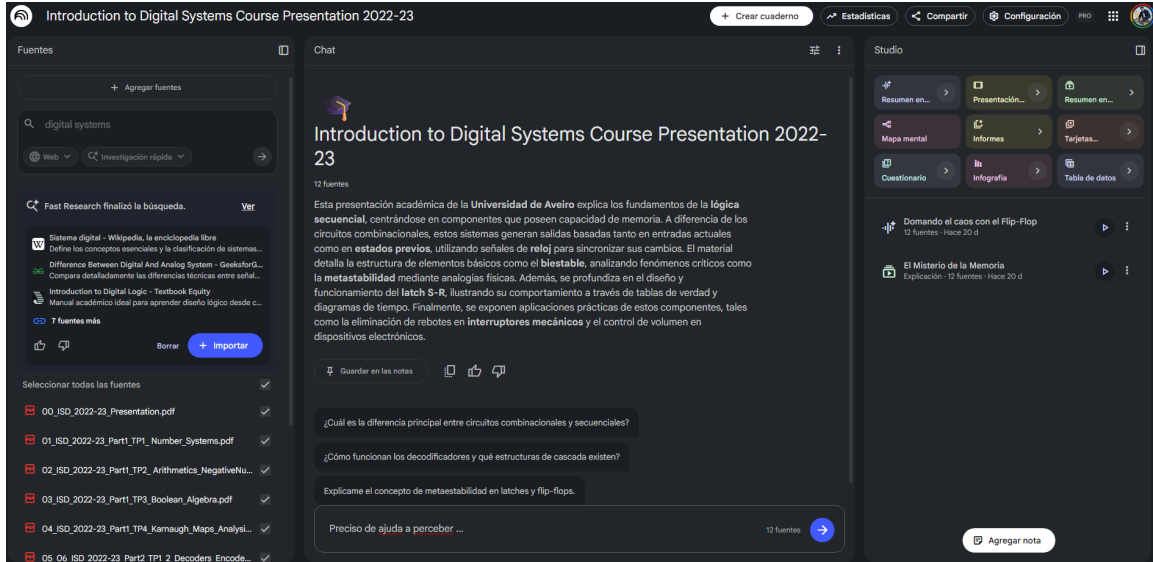


Figure 2.2: NotebookLM interface example, illustrating source-grounded responses and context-aware assistance based on uploaded study materials.

2.1.3 Khan Academy

Khan Academy offers a structured pedagogical framework but remains a predominantly passive learning experience with limited real-time interactivity for complex problem-solving scenarios, particularly in hardware systems. This limitation is supported by Huda et al. [3], which highlights that although students reported high levels of engagement (86% positive perception), learning outcomes were heavily inconsistent. Only a minority demonstrated tangible improvement, while a significant portion experienced performance decline. The study attributes this discrepancy to a lack of adequate scaffolding and insufficient pedagogical integration, indicating that platform interaction alone does not guarantee effective knowledge acquisition. This reinforces the architectural requirement for an adaptive learning approach that incorporates real-time semantic feedback and domain-specific problem-solving support.

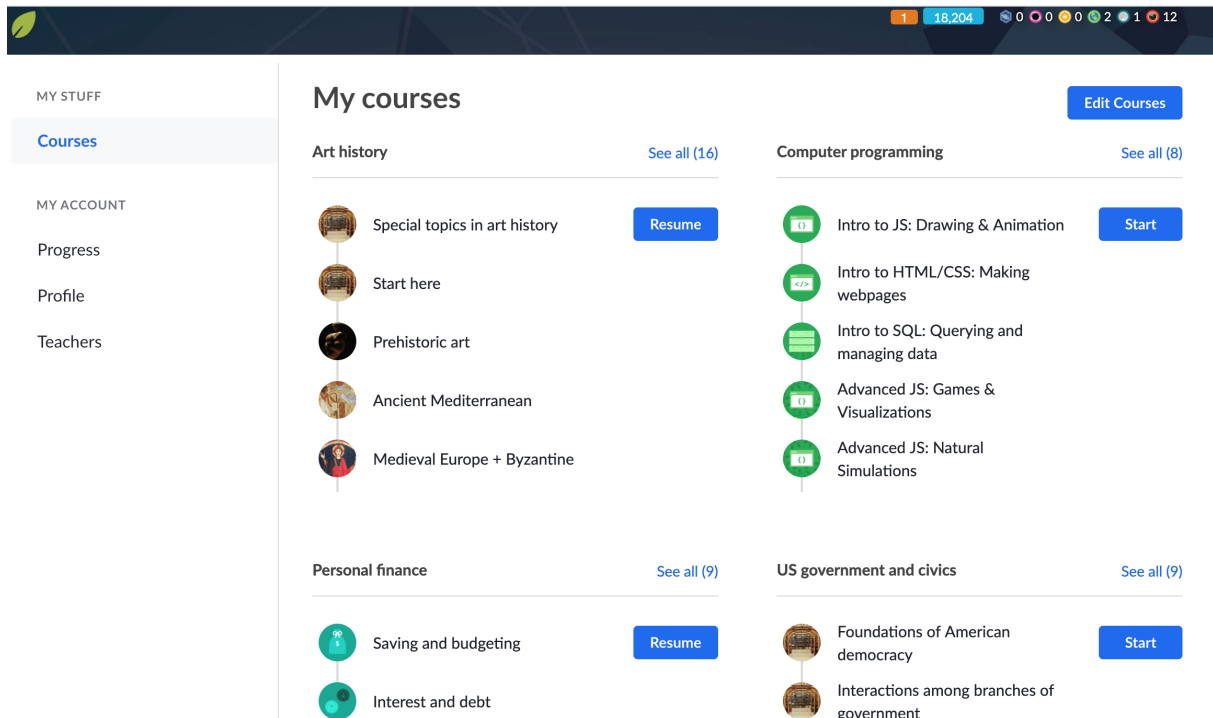


Figure 2.3: Khan Academy interface example

2.1.4 Conclusion

An analysis of the current educational paradigms reveals significant structural gaps when applied to the complex domain of Computer Engineering. While Duolingo excels in user retention through gamification, it operates on static datasets and lacks the procedural depth required for hardware design. NotebookLM demonstrates the power of grounding AI responses in specific literature to prevent hallucinations, yet it remains a passive tool without a structured pedagogical path. Conversely, Khan Academy provides a structured curriculum but offers a predominantly passive learning experience, lacking real-time, semantic bidirectional feedback. The proposed interactive application resolves these deficiencies by converging the strengths of these platforms into a unified, proactive ecosystem.

3. System Requirements and Architecture

3.1 System Requirements

This section presents the system requirements specification, as a result of the first phase of the prototype development. The following subsections begin with a description on the requirements elicitation process, followed by a context description, actors classification, use-case diagrams, non-functional requirements and the system's assumptions and dependencies.

3.1.1 Actors

The target user for the mobile application is a university student, whose prior knowledge of Digital Systems and Computer Architecture may vary significantly. To address this, the interface follows gamification principles to mitigate the steep learning curve of technical subjects. Crucially, the student's learning path is autonomous; the application is designed to function as a self-standing educational tool, allowing students to progress through the curriculum independently of formal instruction. The user of the management interface will typically be a Professor, whose involvement is optional but highly beneficial. While the system operates autonomously for the student, it allows for a Professor to be associated with a specific Course Units (CUs). This enables the instructor to oversee the learning process, ensure the AI-generated content aligns with their specific curricular standards, and monitor student compliance and success rates.

The main actors are described as follows:

- Student: Represents the primary user. The student interacts with the mobile application to complete micro-lessons and challenges. They benefit from an autonomous learning experience where the AI-powered tutor provides immediate feedback. The student does not require a linked Professor to access the core educational features.
- Professor: An optional but specialized actor responsible for pedagogical oversight. This role is designed for monitoring and high-level customization rather than daily system operation.
- Administrator: The technical authority responsible for system health and infrastructure. This actor manages user roles and ensures the platform's security and data protection. They ensure that both the autonomous student experience and the supervised professor environment operate correctly. In case of operational issues, such as removing user accounts or updating CUs, the Administrator is responsible for executing and validating those actions.

3.1.2 Use Cases

The system's functionality is partitioned to ensure a balance between student autonomy and pedagogical oversight. The following table summarizes the primary use cases:

Table 3.1: System Use Cases by Actor

Actor	Primary Use Cases
Student	Consult CUs; Practice exercises;
Professor	Upload CU materials (PDF/Text).
Administrator	Manage users and roles; System health monitoring; Ensure data security and privacy compliance.

3.1.2.1 Student Use Case

The mobile application use case diagram shows the several activities the student can perform on their smartphone to support their learning process. The application focuses on interactivity, real-time feedback, and offline accessibility. The following list describes each use-case:

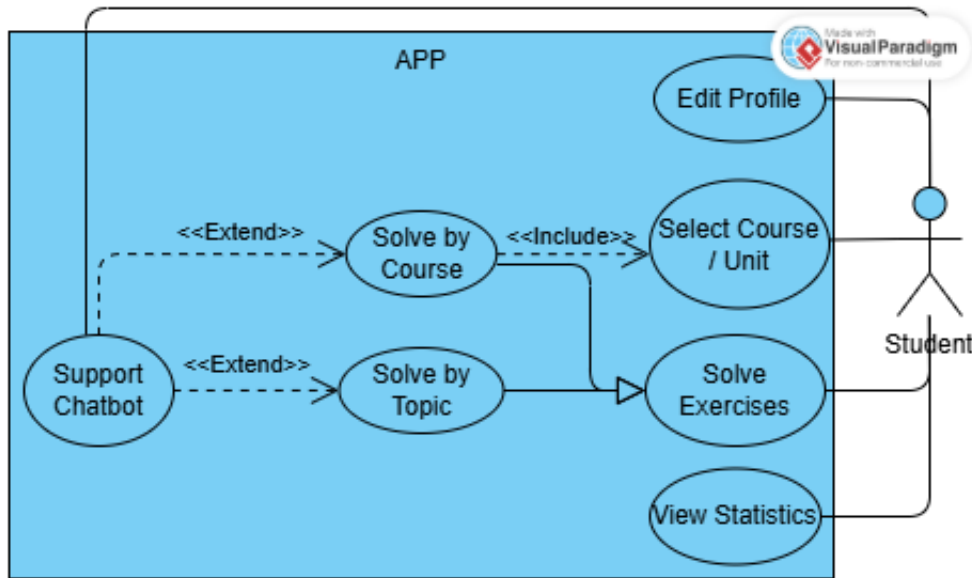


Figure 3.1: Student Use Case Diagram

- **Solve Exercises:** Allows the student to practice Digital Systems and Embedded Systems concepts through questions generated by an LLM. The system presents a series of interactive questions organized by course or topic.
Priority: High
- **View Progress and Performance:** Allows the student to track their learning journey, visualizing earned XP, current level, and knowledge gaps. It includes statistics on accuracy per topic and current "streaks" to encourage daily consistency.
Priority: High
- **Chatbot Support:** Provides an interactive chat interface for students to clarify doubts regarding complex topics. The student sends a query in natural language, and the system utilizes RAG approach to provide accurate answers based on the course's bibliography. The chatbot is only available after the student solve a question.
Priority: High
- **Manage Profile:** Allows the student to customize their learning experience and

personal information. The student can update their avatar, adjust notification settings for daily reminders, and view their achievement badges.

Priority: Low

3.1.2.2 Professor Use Case

The professor interface serves as a management and monitoring tool, allowing educators to oversee class progress and leverage AI to create pedagogical content. The following list describes each use-case:

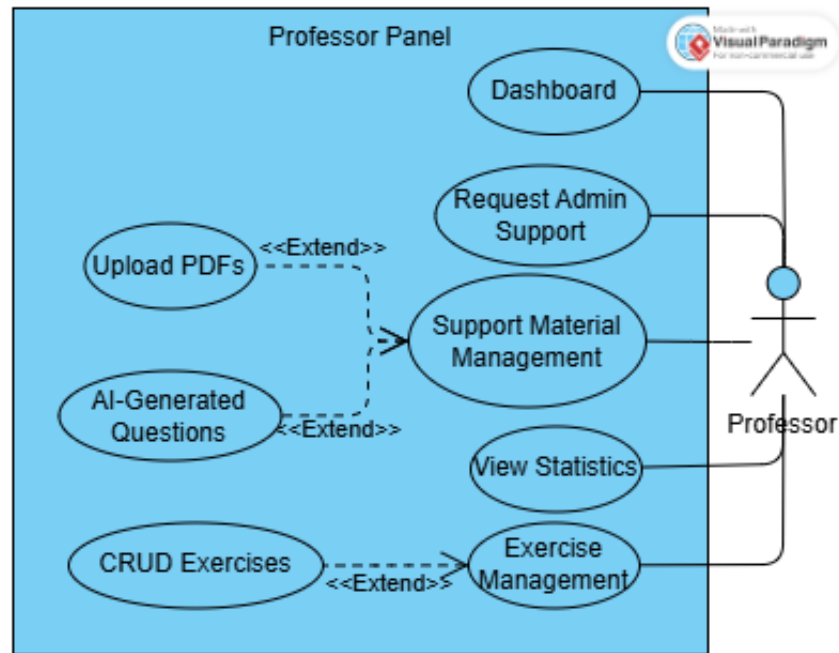


Figure 3.2: Professor Use Case Diagram

- **Manage Content:** Allows the teacher to upload, edit, or remove study materials (PDFs, slides, or lecture notes) that serve as the primary knowledge base for the LLM. This ensures the AI-generated exercises and chatbot assistance are aligned with the specific course curriculum.

Priority: High

- **Generate Exercises:** Provides a specialized interface to prompt the LLM to create new evaluation sets. The professor can define the difficulty level and specific topics. The teacher must review and validate these exercises before they are published to students.

Priority: High

- **Request Support:** Enables the professor to submit formal requests to the system administrator. This includes requesting administrative access to specific CU, reporting technical glitches.

Priority: Low

3.1.2.3 Administrator Use Case

The System Administrator interface is dedicated to the technical management, security, and operational health of the platform. While the professor focuses on pedagogical content, the Administrator ensures that the infrastructure remains robust and compliant.

- **Manage Users & Permissions:** Handles the lifecycle of all user accounts (Students, Professor). This involves approving a new professor account and give per-

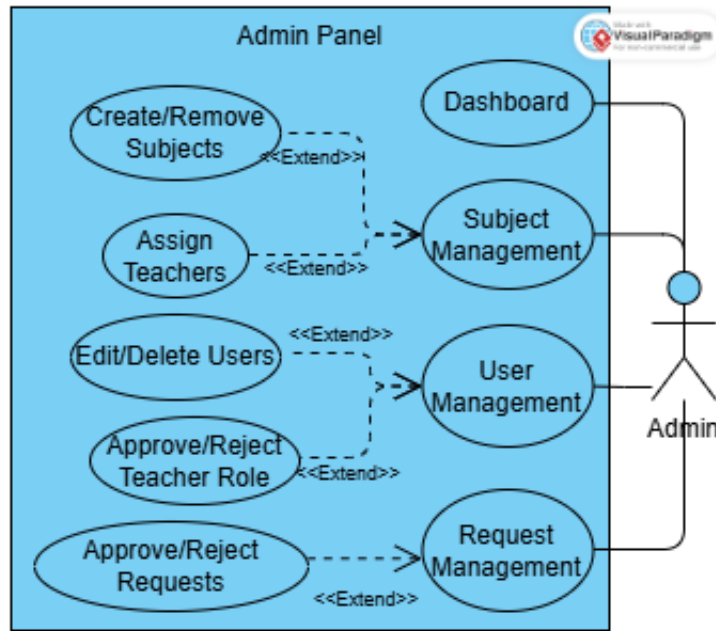


Figure 3.3: Administrator Use Case Diagram

mission to professors to access a specific CU.

Priority: High

- **System Monitoring & Health:** Provides a centralized dashboard to monitor infrastructure activity.

Priority: High

- **Request Management:** Provides a interface that is going to be used to respond and solve professor requests

Priority: High

- **Subject Management:** Responsible for adding and managing CUs.

Priority: High

3.1.3 Non-Functional Requirements

This section outlines the constraints and quality standards the proposed platform must adhere to. These non-functional requirements are essential to ensure the system's scalability, the security of student data, and a smooth interaction flow, directly impacting the long-term adoption of the tool by both students and faculty.

3.1.3.1 Usability

- **Gamified Engagement:** The system shall include clear progression markers, including XP bars, daily streaks, and achievement badges to maintain student motivation.
- **Mobile-First Design:** The UI must be optimized for mobile devices, supporting "micro-learning" sessions that allow students to practice in short intervals.

Priority: High.

3.1.3.2 Performance

- **AI Inference Latency:** To maintain the learner's "flow state," AI-generated feedback and exercise generation should be delivered promptly, ensuring minimal disruption to the learning process.
- **Priority:** High.

3.1.3.3 Security and Data Privacy

- **Secure Communication:** All data exchange between the mobile client, the back-end server, and AI endpoints must be encrypted using industry-standard protocols.
- **Access Control:** A robust authentication module is required to ensure that only authorized students and faculty can access specific class metrics and progress reports.

Priority: High.

4. System Architecture

This chapter describes the architectural framework of the proposed system, detailing the technical decisions and component interactions that support its educational objectives. The architecture is built upon a robust multi-tier design, strategically decoupling user interfaces from the core logic and intelligence engines. By integrating modern frameworks for frontend delivery, a high-performance backend orchestrator, and a state-of-the-art RAG pipeline, the platform ensures a scalable, secure, and pedagogically grounded experience. Figure 4.1 illustrates the deployment model, followed by a detailed analysis of each architectural layer.

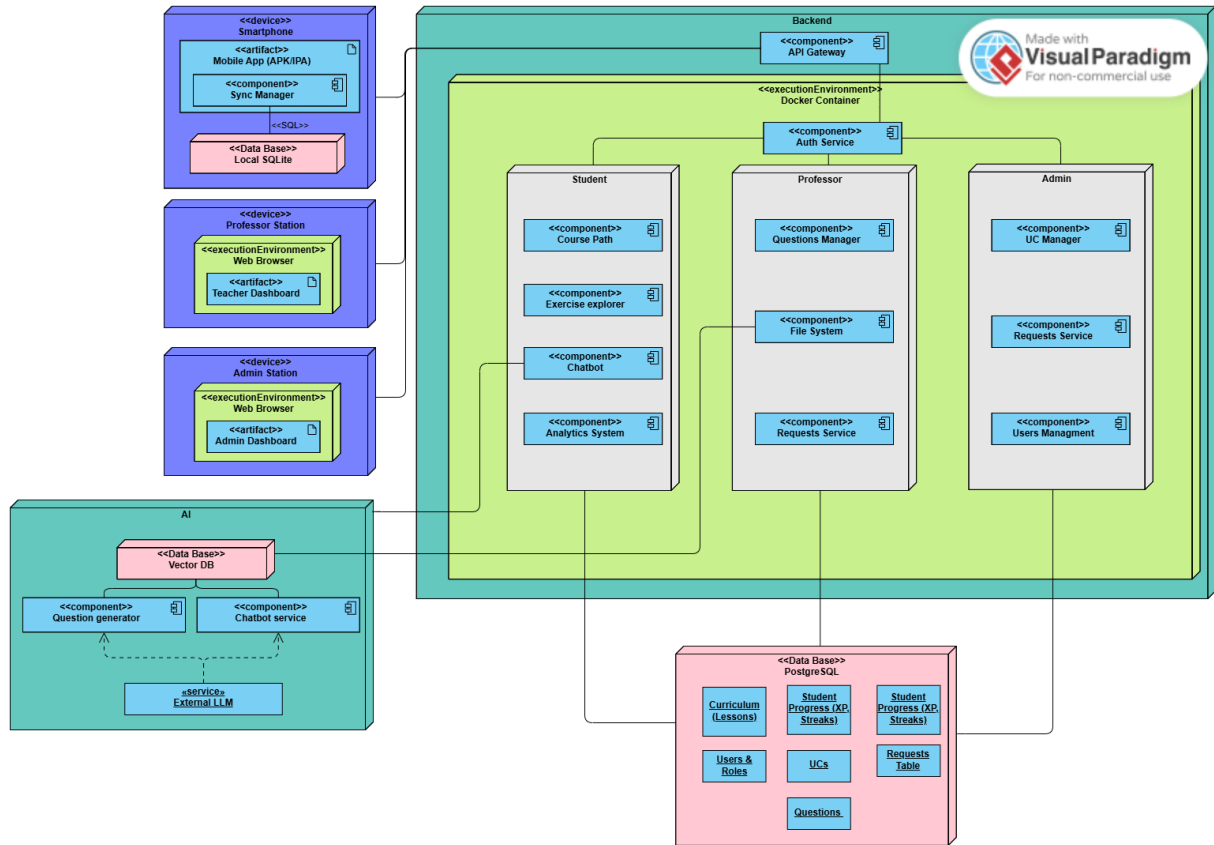


Figure 4.1: System Architecture - Deployment Diagram

4.1 Frontend

The frontend is split into two distinct environments to cater to the specific needs of students and educators:

- **Mobile Application (Student Client):** Developed using the **Flutter** framework. This allows for a high-performance, cross-platform (iOS and Android) experience. Flutter's reactive framework is ideal for implementing the gamified elements (animations, XP bars, and real-time feedback) required to maintain student engagement.
- **Web Dashboard (Professor/Admin):** Built with **Vue.js 3**. This interface focuses on content management, allowing professors to upload pedagogical materials (PDFs/Slides) and monitor class analytics through a responsive and lightweight administrative environment.

4.2 Backend Rationale and Communication

The selection of FastAPI and PostgreSQL is strategic, ensuring high performance and a robust foundation for the educational platform's specialized requirements:

- **High-Performance Asynchronous Processing:** FastAPI's asynchronous nature allows the system to handle multiple concurrent student requests—such as real-time chatbot queries or exercise submissions—without blocking the main execution thread. This is critical for maintaining low response times, a core requirement of the project.
- **Data Integrity for Gamification:** PostgreSQL's ACID compliance ensures that user progress, specifically XP and Streaks, is recorded with absolute consistency. The relational model facilitates complex queries required for the "Statistics Panel," enabling students and teachers to track performance across different Course Units and Topics.
- **Component Interoperability:** The backend acts as the orchestrator between three main pillars:
 1. **Mobile/Frontend Sync:** Handles the synchronization of local exercise caches for the "Offline Mode," ensuring progress made without internet is later validated against the central database.
 2. **AI Service Bridge:** Pre-processes student queries before sending them to the LLM and validates the AI-generated exercises before storing them in the catalog.
 3. **Academic Management:** Links validated teaching materials uploaded by professors to the LLM's context, allowing the "Intelligent Tutoring" system to provide answers based on specific course content.

4.3 AI System

The core innovation of the system lies in its ability to provide grounded, accurate pedagogical support while remaining flexible regarding connectivity. To ensure a clean separation of concerns, the artificial intelligence pipeline operates as a standalone microservice decoupled from the main backend.

- **AI Processing Module:** Developed in Python to leverage its native integration with industry-standard AI frameworks. By utilizing orchestration libraries such as *LangChain*, this isolated module handles the entire workflow: processing the prompt forwarded by the backend, fetching context, and communicating with the models.
- **RAG:** Ensures responses are grounded in academic sources, reducing hallucinations by retrieving relevant context from the course bibliography. This improves both accuracy and efficiency in exercise generation and chatbot interactions.
- **Vector Database (ChromaDB):** Provides fast and lightweight embedding storage and similarity search, enabling efficient retrieval of relevant content while supporting scalable and flexible deployment.
- **LLM Integration:**
 - **Cloud Mode:** Uses an API to access high-parameter models, such as **OpenAI GPT-4o** or **Google Gemini**, for exercise generation and chatbot services.

5. Expected Results

The proposed system is expected to improve student learning outcomes in Digital and Embedded Systems by providing context-aware, adaptive, and interactive educational support. By leveraging Retrieval-Augmented Generation (RAG), the platform should deliver accurate, curriculum-aligned explanations while reducing the misinformation risks typically associated with standalone AI systems.

In addition, integrating gamification with AI-driven content generation is expected to increase student engagement and long-term retention, similarly to platforms such as Duolingo, while addressing their limited technical depth. Compared with traditional e-learning tools such as Khan Academy, the proposed system is expected to provide more interactive and responsive support for problem solving, particularly in complex and domain-specific tasks.

The combination of a vector database (ChromaDB) and large language models should enable efficient real-time feedback and personalized learning experiences, allowing students to progress at their own pace while receiving targeted support. Furthermore, the system is expected to operate effectively in both cloud-connected and constrained environments, ensuring flexibility and accessibility.

Overall, the platform aims to demonstrate that combining grounded AI, interactivity, and domain-specific content can produce higher engagement, stronger conceptual understanding, and more consistent learning outcomes than existing solutions.

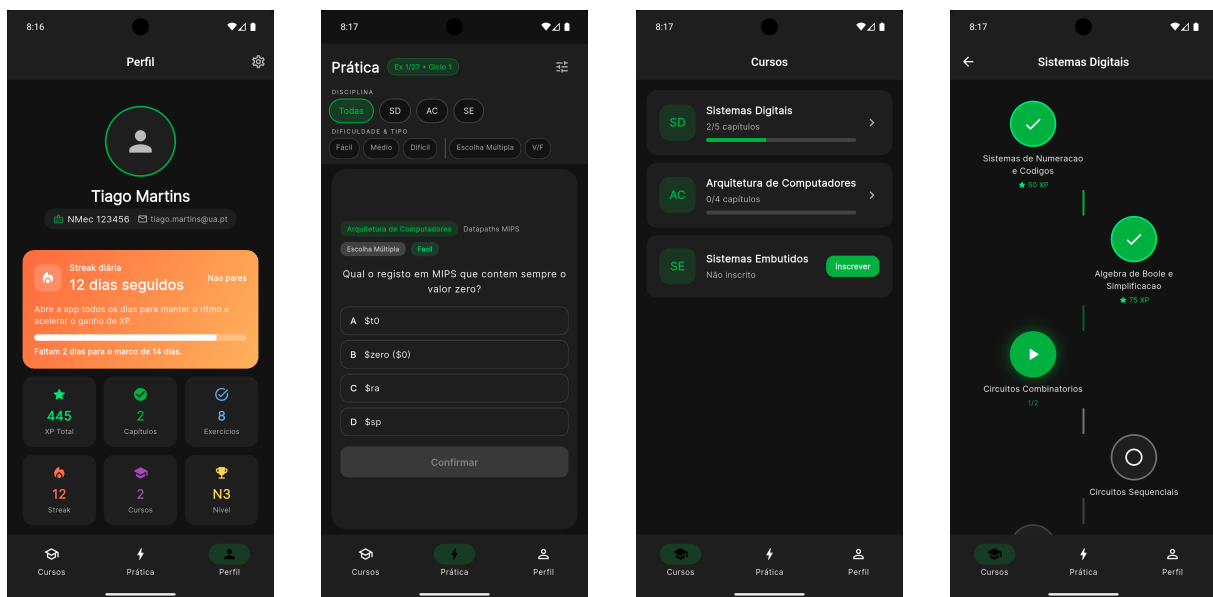


Figure 5.1: Frontend mockups of the mobile application.

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